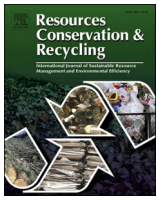
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Full length article



Assessing environmental impacts of nanoscale semi-conductor manufacturing from the life cycle assessment perspective

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ABSTRACT

New products in nanoscale integrated circuit (IC) fabrication for the semiconductor industry are being continuously and rapidly developed. Previous research indicates that the use of energy, water, and other resources by the semiconductor industry has also increased as wafer processing has become more complex and as the industry expands production worldwide, highlighting the environmental impact challenges faced by the semiconductor industry. The industry must re-evaluate its products' environmental impacts and develop new strategies for recently mandated reductions in these impacts. In this research, a study of the impact categories involved in the production of a die of embedded nonvolatile memory (eNVM) was conducted based on life cycle assessment, addressing matters of climate change, water use, particulate matter, and resource use of minerals and metals. The results show that the sustainability of nanoscale IC fabrication of memory products with mask layers has improved from to 2018–2020 based on the midpoint and endpoint results. These preliminary findings support the need for more aggressive strategies for carbon emissions reduction, particularly energy efficiency strategies for SDG 13 Climate Action.

1. Introduction

The greenhouse effect has led to a series of global problems, including extreme weather conditions and increased pressure on businesses to improve environmental performance. To limit global warming by 1.5 °C as required globally, the EU Commission proposed and adopted the European Climate Law in 2021, an agreement that aims to achieve net-zero carbon emissions for EU countries by 2050, mainly by reducing emissions, investing in green technologies, and protecting the natural environment. About a decade ago, Williams et al. (2012) proposed three main energy system transformations to reduce 2050 GHG emissions by 80% in California below the 1990 level: end-use energy consumption, greenhouse gas (GHG) intensity of electricity generation, and electricity share of total end-use energy. Since then, an increasing number of firms have incorporated environmental practices into their strategic plans and operations.

The use of energy and other resources by the semiconductor industry continues to increase as wafer processing becomes more complex and as the industry expands production worldwide. The semiconductor industry accounts for 1.3–2.0% of the total US electricity consumption in the manufacturing sector (Gopalakrishnan et al., 2010). Over time, the semiconductor industry has continued to address climate change and promote environmental sustainability by reducing energy consumption, reducing perfluorinated compound (PFC) emissions, and improving energy efficiency. For example, in 2020, Apple Inc. committed to achieving 100% carbon neutrality for its supply chain and products by 2030. With the changing global climate, TSMC (2021b), a semiconductor manufacturer in Taiwan, has reduced GHG emissions per unit of production [metric ton of carbon dioxide (CO_2) equivalent ($MTCO_2e$) / 8-inch equivalent wafer mask layer] by 40% since 2010. Low-carbon supply chain management has become an important method of achieving sustainable development in the semiconductor industry. This also highlights the industry's response to sustainable development goal (SDG) 13: Climate Action.

In integrating GHG reduction into their energy efficiency strategies, most semiconductor enterprises report on Scope 1 emissions (direct emissions from owned or controlled sources) and Scope 2 emissions (indirect emissions from the generation of purchased electricity, steam, heating, and cooling consumed by the reporting company). Over the past few years, however, attention has shifted to Scope 3 emissions

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(emissions from the corporate value chain) because the largest proportion of GHG emissions accrues upstream and downstream in the value chain. In this effort, corporations must obtain information from other firms, institutions, and consumers outside the reporting company's control Patchell (2018). observed that Scope 3 emissions reporting is a complicated challenge for most enterprises. This is especially true for more complex products such as semiconductors. It is extremely difficult to allocate CO_2 emissions from the supply chain in such a way that a customer, such as Intel, can estimate its portion of the impact from the manufacture of a material that is also used by other customers (Higgs et al., 2009).

The management and assessment of hazardous chemicals are complicated by embedded trade secrets and continual rapid changes in the electronics manufacturing process (Mullen and Morris, 2021). In the digital era, new products are being developed more quickly and their prices are falling. Semiconductor devices involve a complex set of nanoscale fabrication processes that are energy -and resource-intensive and generate significant waste (Krishnan et al., 2008). During the IC manufacturing processes, which involve a diverse range of chemicals and processes combined in a complex workflow (Boyd, 2012), significant amounts of water are used and further CO_2 emissions are generated it. With the shift to global net-zero carbon emissions, companies are under increasing pressure to improve their environmental performance, especially in the semiconductor industry. Net-zero carbon emissions are achieved when there is a balance between the carbon emissions produced and those removed from the atmosphere.

Few prior studies have discussed this challenge, highlighting the following research gaps. First, is the semiconductor industry a high-risk environmental impact industry? Second, if it is, has the semiconductor industry become more sustainable? To bridge this research gap, this study conducted a systematic life cycle assessment (LCA) involving the analysis of different nanoscale processes; a case study involving a memory manufacturer was conducted to illustrate the results.

The remainder of this paper is organized as follows. A literature review is presented in Section 2 Section 3. describes LCA studies on semiconductors, and Section 4 provides the results. Finally, the conclusions and opportunities for future studies are presented in Section 5.

2. Environmental impacts of the semiconductor sector

The demand for integrated circuit (IC) components has increased with the ever-expanding use of digitalization and artificial intelligence (AI). The IC is a core component that enables device functioning. The four types of IC chips are memory IC, logic IC, analog IC, and micro IC. The rapid increase in chip production has had considerable environmental effects during different phases of their life cycle (Villard et al., 2015).

Within advanced nanoscale process technology, chip sizes have been investigated and developed for different applications in small smartphones, portable devices, and the automotive industry. Moore's Law (1965) is a famous golden rule in the field of electronics. Gordon Moore, one of Intel's co-founders, observed that as the number of transistors on a microchip increases rapidly, exponentially increasing computing power, the chip cost decreases. Consequently, the foundry process technologies become more complex Fig. 1. shows the evolution of foundry process technologies over the last four decades (TSMC 2021a). With increasing complexity, a silicon disk can take three to four months to pass through the multiple processsing stages to a finished product.

Foundry and memory manufacturing are two major business segments in IC fabrication. Wafer fabrication involves a complex process within the semiconductor manufacturing process. It is a resource-intensive process conducted in an antiseptic room using a substantial amount of chemicals, gasses, energy, and water (Ahmad, 2007). The manufacture of many modern semiconductor devices involves a complex set of energy-intensive and resource-intensive fabrication processes that generate significant waste (Krishnan et al., 2008). Previous research has shown that semiconductor devices consume large amounts of materials, chemicals, energy, and water throughout their life cycle, particularly during the manufacturing phase (Kuo et al., 2020).

The environmental impacts of industrial activities can be characterized using two methodologies: a top-down methodology based on economic input-output (EOI) analysis, or EIO-based LCA (Virtanen et al., 2011; Wang et al., 2020) and a bottom-up methodology based on process analysis, known as process-based LCA (Kuo, 2013; Kuo et al., 2014; Zhang et al., 2019). EIO is a useful tool for determining sectoral impacts, whereas process-based LCA is used to measure the impact of individual products from cradle-to-grave or cradle-to-cradle (Mattila, 2018). These two LCA categories are the most commonly used.

LCA is useful for estimating the environmental impacts of products and services, determining hotspots, and guiding the development of sustainable products or services. Additionally, LCA provides a comprehensive evaluation of all upstream and downstream energy, resource inputs, and environmental emission outputs (Kuo et al., 2014). LCA requires considerable time and effort to collect raw data. Allocation rules must be determined to properly conduct an LCA when the raw data of the research subject cannot be clearly obtained, according to Huang et al. (2016).

Life cycle impact assessment (LCIA) is used to evaluate potential environmental impacts by converting the life cycle inventory (LCI) results into specific impact indicators. There are many different impact assessment methods for calculating the LCIA results. Among these, ReCiPe is an LCIA method that follows the midpoint method of Chain Management by LCA Impact Assessment (CML-IA) and the endpoint method of Eco-indicator 99 (Goedkoop et al., 2009). ReCiPe can be divided into two main approaches to characterization, namely midpoint and endpoint (Dong and Ng, 2015). The endpoint LCIA estimates the damage to the areas of protection (i.e., human health, ecosystems, and resource depletion), whereas the midpoint assessment intends to analyze the amount of emissions of various impact categories. These two approaches are complementary because of the different levels of uncertainty in the characterization factors.

There are very few published LCA-related case studies from an industrial initiative perspective (Kuo et al., 2020; Liu et al., 2010), especially in the semiconductor industry, for several reasons. First, the assessment of the impacts of a microelectronic chip alone is insufficient to effectively draw the profile of this complex industrial sector. Second, nanoscale fabrication processes change rapidly, as observed in Moore's Law. The product tends to expire rapidly, requiring more components per wafer (units of production) to be embedded, which separates the various levels of advancing development across time, for example, from 20 nm to 30 nm to 40 nm. This suggests that the data reliability of previous studies is too low for electronics. These characteristics hinder LCA in the semiconductor industry (Liu et al., 2010).

Boyd (2012) presents a comprehensive LCA of semiconductor devices and electronics. However, the rapid increase in chip production has considerable effects on the environment during different life-cycle phases Villard et al. (2015). pointed out that microelectronics analyses lag in the environmental field Table 1. shows LCA-related studies in the

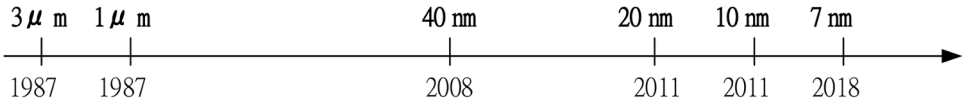


Fig. 1. Evolution of nanoscale fabrication processes (TSMC, 2021).

Table 1
LCA related studies on semiconductor industry.

Authors (date)	Phase	Functional unit	Method	Results
Ahmad (2007)	CMOS wafer	200 mm wafer	process	It showed that approximately 75% of electricity usage contribute to the environmental burden associated with CMOS wafer manufacturing.
Krishnan et al. (2004)	nanoscale fabrication processes	8 layers of copper wiring, manufactured on a 300 mm wafer	EIO	The results could be used to estimate and improve the environmental performance of a broad range of electronics and other emerging applications that involve nano and micro fabrication
Higgs et al. (2009)	scope 3	Intel processor	N.A.	The relative CO_2 impacts of the various aspects of semiconductors, including manufacturing operations, product use, and various scope 3 impacts, such as supply chain, logistics and employee travel are illustrated.
Liu et al. (2010)	DRAM production	8-inch wafer	processes	The impacts of five production processes in the DRAM manufacturer are discussed.
Boyd et al. (2010)	CMOS transistor	a die	processes	A LCA for digital logic chips over seven technology generations, spanning from 1995 through 2010 is provided.
Andrae and Andersen (2011)	packaging	BGA/CSP process	processes	Different packaging methods are evaluated based on the LCA method.
Huang et al. (2012)	function processes	a wafer	regression model	Several regression models were developed based on the product characteristics (function, technology node, mask layer, and metal layer).
Boyd (2012)	DRAM	N.A.	process	The LCA of semiconductor devices and electronics is explained.
Wang (2014)	IC foundry	N.A.	regression model	The major contribution of carbon footprint of the IC foundry is the electricity consumption (64%) in the manufacturing phase is explained.
Wu (2014)	packaging	N.A.	process	The mask layer and etched layer were positively correlated to the carbon footprint, while the process technology was negative correlated to the carbon footprint.
Huang et al. (2016)	Per wafer	a wafer	regression model	The mask layer could be the most important parameter for predicting CFP of wafer because of its higher standardized coefficients in each regression model.
Kuo et al. (2020)	packaging	packaging process	process	Two commonly used materials, gold and copper wires, are analyzed. The environmental damage using a comprehensive approach (ReCiPe 2016)

Note:.
1. CMOS (complementary metal oxide semiconductor),.
2. DRAM (dynamic random access memory).
3. N.A.: not available.

semiconductor industry. Most research defines a wafer or a die as the functional unit (Boyd et al., 2010; Huang et al., 2012, 2016; Liu et al., 2010) and the mask layer is regarded as the most significant parameter for the difference of environmental impacts (Huang et al., 2016; Wu, 2014) Table 1. shows LCA-related studies on the semiconductor industry.

Villard et al. (2015) identified key indicators of the direct and indirect environmental impacts of the semiconductor industry. These impacts include global warming, abiotic depletion, water consumption, summer smog, and local electricity consumption Table 2. shows the significant environmental impacts of the semiconductor industry extracted from that work.

Semiconductors are crucial components in millions of electronic devices in the digital world. Their environmental impacts are significant, particularly GHG emmissions and water use. In the literature review, we found that most of the studies on semiconductor environments were conducted in 2010. These studies are essentially obsolete, as the industry is rapidly changing. Thus, a contemporary analysis of the environmental impacts of semiconductors and a comparison of the environmental impacts of advanced technologies in the semiconductor industry are deemed necessary.

Table 2
Significant environmental impacts of the semiconductor industry (Villard et al., 2015).

Indicators	Relevance
Global warming	It is crucial that a huge amount of electricity is consumed throughout the life cycle.
Abiotic depletion	Chip manufacturing consumes energy and mineral resources.
Water	The quality of water surrounding microelectronic plants is damaged by the intensive usage of nitrogen and phosphorous acids, especially during wet cleaning processes.
Summer smog	Several steps of wafer and package processing consume solvents that produce VOCs (volatile organic compounds) and plant facilities damage the air quality.
Electricity consumption	This indicator accounts for the total energy consumed by equipment and facilities during manufacturing.

3. Research method

In this study, we conducted an LCA focusing on the semiconductor industry based on the ISO 14,040 LCA series standards, ISO 14,067 carbon footprint standards, and product category rules (CFP-PCR, 2012; PCR, 2011). Wee investigated the system boundary and functional unit factors, LCI, data collection, and LCIAs. In addition, we followed the product category rule (PCR) for DRAM.

3.1. System boundary and functional unit

The goal definition process establishes the breadth, depth, and scope of a system and establishes the appropriate types of data required for the assessment.

Memory production involves a series of complex manufacturing processes completed within a two-month period. Although different facilities may have slightly different groupings, the five processes to be considered in the analysis are diffusion, photolithography, etching, thin film, and chemical-mechanical polishing (CMP).

Two major types of production operations are employed in semiconductor manufacturing. The first takes place in a fabrication plant (typically referred to as a “fab”) where individual computer chips are made on silicon wafers. At this point, all the circuitry needed to make a functioning computer chip is complete, but the chip is not in its final packaged form. The second major manufacturing operation involves assembly and testing. In this operation, the silicon wafer is cut into individual chips, and the chips are placed in the final package, which enables them to be inserted into a computer or another finished product. The fab portion of the operation requires several chemical processes, gas inputs, and raw silicon.

There are several definitions of functional units Liu et al. (2010). defined a functional unit as “a wafer.” Huang et al. (2016) defined functional units as functional types. In this research, a functional unit is defined as “a die for the technology node and mask layer” as a high yield rate is considered. Normally, the system boundary includes DRAM production and testing. However, since the test process is outsourced in

the selected case, it was excluded from the inventory system (Fig. 2).

In the semiconductor industry, silicon is the most important substrate material in the raw material stage. The manufacturing process occurs as follows: First, during diffusion, a doping agent (dopant) is added to the semiconductor substrate to alter its conductivity. Secondly, during the photolithography process, the designed pattern from the mask or reticle is transferred to the photoresist on the wafer surface. Third, during the thin-film process, metal layers are deposited onto the wafer surface. Fourth, during the etching process, the doped or undoped oxide layers are removed from the entire wafer. Finally, during CMP or planarization, a combination of chemical and mechanical forces is used to smooth the surfaces. CMP is a hybrid of chemical and free-abrasive polishing methods.

3.2. Life cycle inventory and data collection

LCI development is a data-based process for quantifying inputs and outputs throughout the lifecycle of a product, process, or activity. Inputs refer to raw materials and energy, whereas outputs are products including solid, liquid, and gaseous emissions. Inventory data were obtained from questionnaires in a memory manufacturing industry that required the ISO 14,064 standard Scope 1 (direct emissions from owned or controlled sources) and Scope 2 (indirect emissions from the reporting company, including consumption of purchased electricity, steam, heating, and cooling).

In this study, the environmental impacts of different semiconductor technology nodes were analyzed. A technology node is defined according to the ground rules of a process governed by the smallest feature printed in a repetitive array (Huang et al., 2012). Technology nodes of 40 nm, 30 nm, and20 nm specifically refer to the generation of chips made using a particular technology and differentiated by size. Data were collected over three years from January 2018 to December 2020.

All the energy and materials that were transported into the system, together with the discharge of environmental pollutants, were calculated using technology nodes. Referring to the memory product category rule, inventory items at each stage of the LCA were as follows:

(1) The raw material acquisition stage, including: (1) materials that constitute components of, accessories for, and packing for the target product, and (2) transportation of raw materials, auxiliary materials, and energy resources to the target factory.

(2) The manufacturing stage, including: (1) energy resources used in the manufacturing process of the target product; (2) pure water manufacturing, wastewater and waste gas disposal, and refrigerants; and (3) removal and disposal of air pollution, water pollution (including any illegal discharge of controlled items), and waste resulting from the manufacturing of the target products.

Suitable allocation rules should be identified when conducting semiconductor product LCA. The principles of physical allocation can be divided into three methods: (1) the marginal variation method, (2) a representative parameter method that reflects the common function of the products, and (3) the causal physical relation allocation method, applicable when there is a physical indicator of the cause-and-effect relationship between the products and associated emissions or resources used (Jolliet et al., 2015). In this research the "number of masks" was used as a benchmark for the allocation ratio (Huang et al., 2016). This means that there are more masks used for 20 nm than for 40 nm. All semiconductor devices are manufactured as a series of mask layers on a substrate material, and each layer is created from a different mask and material. Three different methods were used for transportation.

(1) Land transportation calculation: Google Maps
(2) Air transportation: International Civil Aviation Organization (ICAO) Carbon Emissions Calculator (ICAO, 2021)
(3) Sea transportation: Ports.com−sea route and distance (ST, 2021)

Several assumptions were made based on interviews with memory plant engineers. The number of wafers for the target products (p_{ax_i}) was calculated. The number of masks was used as a benchmark to allocate the consumption of total electricity, other fuels, chemicals, and waste use as per Eqs. (1) and (2):

$$\begin{aligned} &Numbers\ of\ mask(m_{ax_i})\ for\ the\ target\ product \\ &\quad = number\ of\ masks\ per\ wafer\ (q_{ax_i}) \\ &\qquad \times\ total\ number\ of\ wafers\ for\ the\ target\ product(p_{ax_i}) \end{aligned} \tag{1}$$

$$Allocation\ ratio\ (a) = \frac{m_{ax_i}}{\sum_{j=1}^{n} m_{ax_i}},\ Where\ m_{ax_i} = q_{ax_i} \times p_{ax_i} \tag{2}$$

All purchased item data were directly downloaded from Enterprise

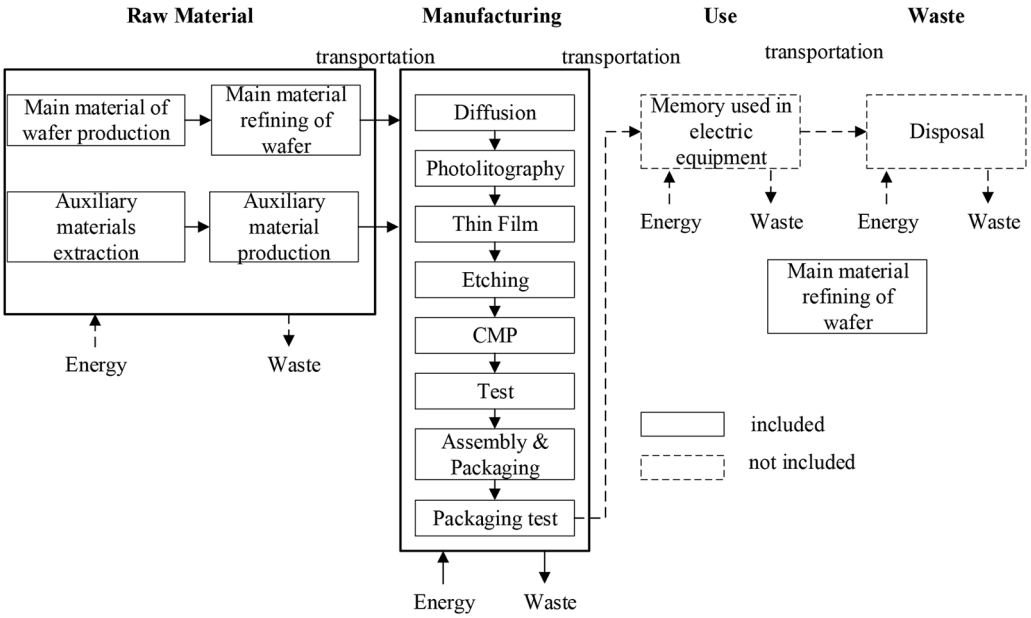


Fig. 2. System boundary of this research [adapted from Liu et al. (2010); (PTC, 2010)].

Resource Planning (ERP). For certain chemicals, such as photo-resistive liquids, slurries, and developers, we used information from material safety data sheets (MSDS) to calculate the mass and weight proportion by mass balance from chemical equations.

In the LCIA, the environmental impacts of each target product were based on inventory data (electricity, fossil fuel, and some factory-wide auxiliary materials).

3.3. Life cycle impact assessment

The LCIA evaluates the actual and potential environmental effects of environmental loadings or resource requirements owing to the inputs and outputs of the inventory components. SimaPro 9.0 was used to evaluate the environmental impacts of the semiconductor foundry process. The ReCiPe 2016 (GWP100; IPCC2007) evaluation method, which provides midpoint and endpoint impact scores, was used to calculate the environmental impact at different stages in the cause-effect chain. Midpoint indicators focus on a single environmental problem, such as climate change or acidification. Endpoint indicators, however, show the environmental impact at three higher aggregation levels: (1) effect on human health, (2) biodiversity, and (3) resource scarcity (Bare et al., 2000).

Based on Villard et al. (2015), global warming, abiotic depletion, water, summer smog, and local electricity consumption should be considered in LCIA of the semiconductor industry. Global warming is related to climate change and energy use, abiotic depletion is related to energy and mineral resources, and summer smog is related to resource use and particulate matter (PM). Therefore, four environmental impacts were investigated in this study: climate change, water use, PM, and mineral and metal resource use.

(1) Climate change. Net-zero CO_2 [CO_2 equivalent (CO_2eq)] is targeted for 2050. The global warming potential (GWP) of all GHG emissions was compared to the GWP of 1 kg of CO_2. A GWP of 100 (GWP 100) was used to calculate CO_2 equivalence. GWP compares the global warming impacts of different gasses. Specifically, this is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of CO_2 (EPA, 2021). The GWP of CO_2 is 1.
(2) Water use Hoekstra (2016). stated that water is a global resource because it is virtually traded via products as well as between water-abundant and water-scarce regions. The average precipitation in Taiwan is approximately 2500 mm. From 1950 to 2021, the precipitation was a fraction of 2300 nm. The impact category considers the availability or scarcity of water in the regions where the activity takes place, if this information is available. Unit of measurement: cubic meters (m^3) of water use related to the local scarcity of water.
(3) PM. The size of particles is directly linked to their potential to cause health problems. PM is further distinguished according to aerodynamic diameter: respirable particles $(PM_{10}) < 10\mu m$; fine particles $(PM_{2.5}) < 2.5\mu m$; and ultrafine particles $(UFPs) < 100\mu m$ aerodynamic diameters (WHO, 2006). $PM_{2.5}$, which is recommended for the LCIA, may best describe the components of PM responsible for adverse health effects (Fantke et al., 2015; Harrison and Yin, 2000; Lippmann and Chen, 2009).
(4) Mineral and metal resource use. The use of mineral resources in a product system can affect future users' access to these resources. Two important elements are related to resource use: potential availability and accessibility (Berger et al., 2020). The rate of resource depletion was converted into equivalents of kilograms of antimony. Unit of measurement: kilogram of antimony equivalent (kg Sb eq.)

The uncertainties associated with environmental impacts can be categorized into scientific uncertainty and estimation uncertainty. Estimation uncertainty can be further classified into two types: model and parameter uncertainty.

4. Results and discussion

In this study, a real-life IC memory fabrication process was investigated. The company profile is omitted here because the data are confidential. Inventory data were collected from the two factories. An LCA was conducted for all products. Three different technologies were used in the foundry process: 20, 30, and 40 nm. The inventory data for the energy and resource inputs included electricity, diesel, natural gas, tap water, chemical gas, and common acids, which are generally industrial grade, with a purity of 99% or lower, rather than ultra-high-purity semiconductor grade (99.9997% – 99.999999% pure) (Boyd, 2012; Huang et al., 2016). The output of the middle point was represented by waste emissions, including waste air, wastewater, and solid waste. A sensitivity analysis of the ReCiPe 2016 endpoints was conducted.

4.1. Analysis of environmental impacts of different generation products

Based on Villard et al. (2015), four significant environmental impacts for the semiconductor industry were selected for this research: (1) climate change, (2) water resources, (3) PM, and (4) resource use, including minerals and metals.

(1) Climate change

Fig. 3 shows the average GHG emissions for the different technologies and products. The GHG emissions of the 40 nm type technology were the highest. Notably, CO_2 emissions were reduced from 2018 to 2020.

The GHG emissions of the 20 nm and 40 nm technologies were reduced by −75% in 2018. The GHG emission ratios were reduced by −24% (20 nm), −33% (30 nm), and −46% (40 nm) for 2018–2020.

(1) Water resources

Fig. 4 illustrates the water resource data for the technologies and products. The water resources required by the 40 nm technology are the highest.

In 2018, the water resources for the 20 nm and 40 nm technologies were reduced by −76%. The water resource ratios were −65% (20 nm), 81% (30 nm), and 78% (40 nm) for 2018–2020, respectively.

(1) Particulate Matter

Fig. 5 shows the PM in the different technologies and products. The PM of the 40 nm type technology is the highest.

The PM of the 20 nm technology and 40 nm technology was reduced by −74% in 2018. The PM ratios were −40% (20 nm), −44% (30 nm),

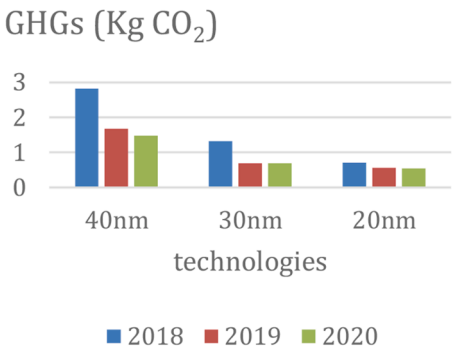


Fig. 3. Average GHG emissions, 20 nm, 30 nm, and 40 nm.

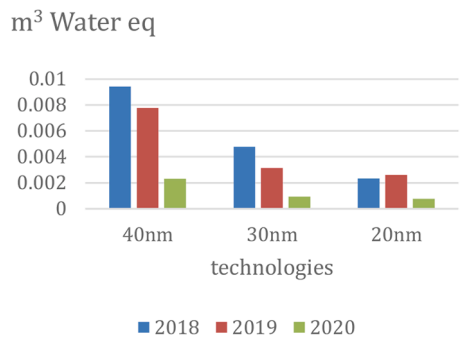


Fig. 4. Average water resources, 20 nm, 30 nm, and 40 nm.

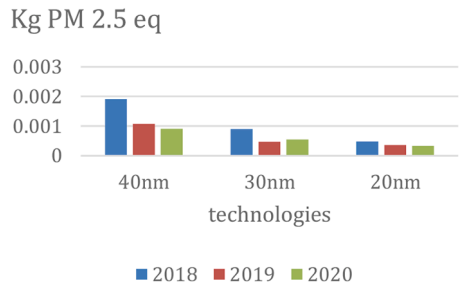


Fig. 5. Average PM of 20 nm, 30 nm, and 40 nm.

and −53% (40 nm) for 2018, 2019, and 2020, respectively.

(1) Resource use, including minerals and metals

Fig. 6 shows the mineral and metal resource use in the different technologies and products. The 40 nm technology had the highest mineral and metal resource use.

The resource use of minerals and metals in 20 nm and 40 nm technologies was reduced by −75% in 2018. The resource use ratios were −24% (20 nm), −33% (30 nm), and −46% (40 nm) for 2018, 2019, and 2020, respectively.

Finally, a sensitivity analysis of the uncertainty endpoints was conducted. In this study, the uncertainty was assumed as 95%, 100%, and 105%. In addition, the highest sales volume was selected for each new technology node. From Figs. 7-9, it can be seen that the damage to human health was greater than the damage to ecosystems or resource availability. These results also show that the environmental impacts of the 20 nm and 30 nm products were less than for the 40 nm. Additionally, the impact of 40 nm was significantly reduced from 2018 to 2020.

The unit for human health damage refers to the disability adjusted life years lost in the human population; the units for ecosystem damage refer to the number of species lost integrated over time; fossil resource scarcity is the only midpoint category which do not have a constant midpoint to endpoint factor.

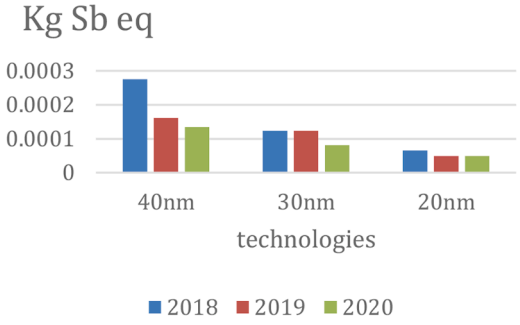


Fig. 6. Average resource use: minerals and metals of 20 nm, 30 nm, and 40 nm.

4.2. Environmental impacts for different life cycle stages

Throughout the life cycle stages, 85% of climate change and PM contributions occurred during the manufacturing stage. In contrast, most impacts on water resources, minerals, and metals originated during the raw-material stage Fig. 10. shows the significant environmental impacts of each stage. Normally, the environmental impacts of raw materials originate from the supply chain, as Scope 3 emissions (value chain emissions). In this case, the environmental impacts of manufacturing were from the manufacturing site, categorized as Scope 1 and 2 emissions. This highlights the need for a new approach to emission reduction strategies that involve semiconductors.

Results of the analysis of the contributions of environmental impacts across the various materials and chemicals are shown in Table 3. According to the analysis, the two most important inventory materials are electricity consumption (manufacturing stage) and wafers (raw material stage). The results of the case study involve the reduction of H_2O_2 and $CaCl_2$.

5. Conclusions

Semiconductor products are important owing to the magnitude and exponential growth of their production. The effects of climate change and extreme weather events such as droughts and storms are increasingly common, with threats to the electricity and water supply in the semiconductor industry. Manufacturing processes that transform wafers into hundreds of individual dies require the use of chemicals and materials such as acids, bases, and solvents for deposition, patterning, selective removal, and cleaning. Rare minerals, such as tin, tungsten, tantalum, and gold must be mined for use in the semiconductor industry.

Several observations can be made from the analysis of the IC memory foundry process. First, the environmental impact of the advanced nanoscale technology of 20 nm is less than that of 30 and 40 nm. Second, the most significant environmental impacts originate from electricity consumption (during the manufacturing stage) and the use of wafers (at the raw material stage). Third, products reliant on semiconductors are diverse and include personal computers, servers/industrial computers, TV/set-top boxes, blue ray players, telecommunications, digital cameras, gaming consoles, smart phones, terminal vending machines, printers, and tablets. Among these, telecommunication devices generate more CO_2 emissions due to the higher number of mask layers. Therefore, engineering designers should emphasize the energy efficiency of telecom products to reduce CO_2 emissions.

5.1. Research contributions

Semiconductor enterprises have implemented projects to reduce GHG emissions and water use. The evaluation of the reduction baseline for semiconductor companies is a challenging but effective starting point for impact mitigation. Traditionally, the inventory process of LCA in the semiconductor industry has not been easy to perform. Previous research has focused on modeling for estimation. This study bridges these research gaps. The contributions of this study are as follows: First, we selected the critical factors affecting the environmental footprints of the selected functional unit, the “die for technology node and mask layer”, namely climate change, water use, PM, and minerals and metals. We analyzed these factors using LCA for a recent timeframe spanning 2018 – 2020. In addition, different nanoscale foundry process technology nodes and mask layers were compared. Second, according to Moore’s Law, the number of transistors on a microchip doubles every two years, while the cost of the microchip is halved. Therefore, the environmental impact was halved during the timeframe of the study. Third, most of the environmental impacts are caused by electricity consumption and wafer use.

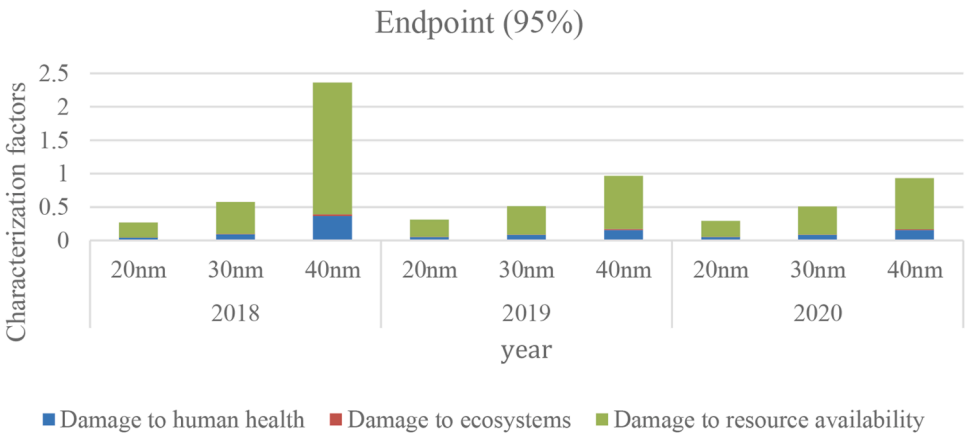


Fig. 7. Endpoints of data activities at 95% uncertainty.

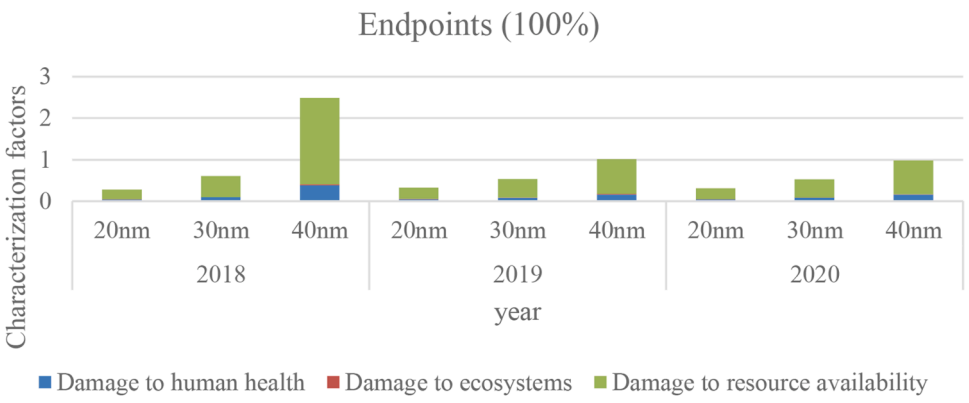


Fig. 8. Endpoints of data activities at 100% uncertainty.

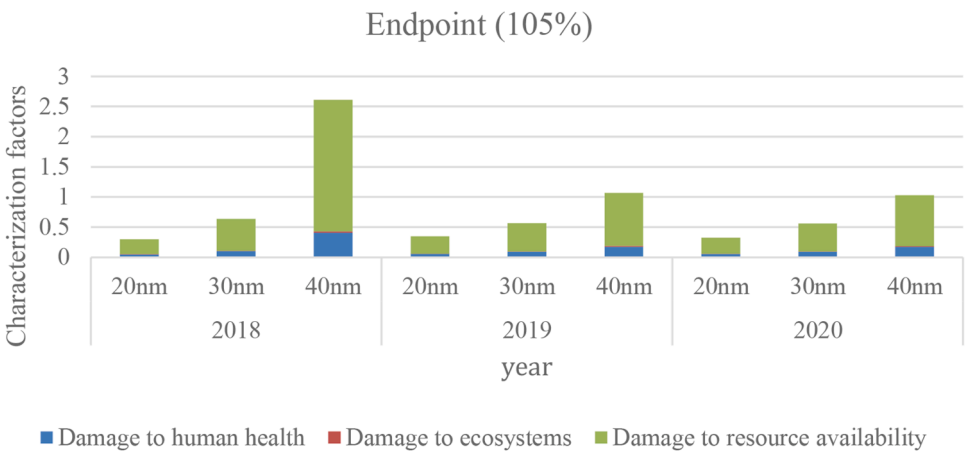


Fig. 9. Endpoints of data activities at 105% uncertainty.

In this study, approximately 80% of GHG emissions originated from electricity use (Scope 2) is. Fourth, the CO_2 levels for product use were calculated. The results show that most CO_2 emissions are from telecommunication device use. Productivity and yield rate are two key factors related to CO_2 emissions. A 20 nm technology node was introduced in the market at a time when the production volume was lower and the yield rate was higher.

From a managerial perspective, regulation of electricity consumption and water use should be tied to national policy. This is directly related to cleaner energy investment. Semiconductor companies should begin implementing short-term projects to reduce and stabilize their electricity consumption and water use. Advanced technology is cost-effective and has fewer adverse environmental effects. Currently, the EU is considering the introduction of a carbon border adjustment mechanism (CBAM) for all imports to be in place no later than 2023. Policies on carbon tax (Kuo et al., 2016) and carbon leakage (Yu et al., 2021) will have a significant impact on carbon reduction in the semiconductor industry. Therefore, innovative projects based on advanced technology should be incentivized to assist the semiconductor industry in achieving net-zero emissions. Semiconductor companies should

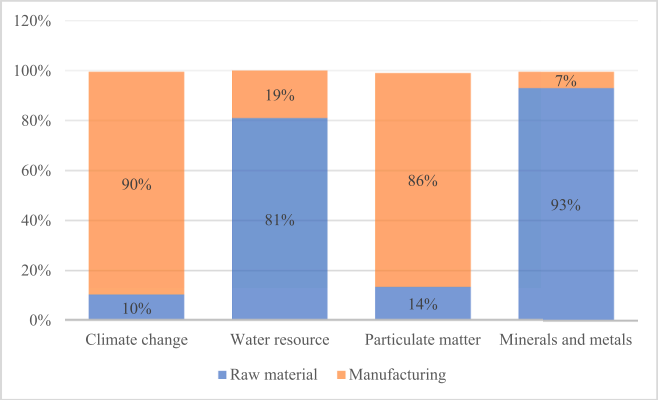


Fig. 10. % contribution to environmental impacts in different stages.

Table 3
Material/chemical hotspots in relation to environmental impacts.

Environmental impacts	First hotspot	Second hotspot
Climate change	Electricity	Wafer
Water resource	Wafer	Water
Particulate matter	Electricity	Wafer
Minerals and metals	Ta	H_2SO_4

consider advanced technologies for carbon emission capture, utilization, and storage.

5.2. Limitations and future studies

This study has some limitations. First, the data collection and analysis was limited to one case study. In the future, data could be collected from multiple companies to obtain a more robust database. Second, in this study, the scope of the life-cycle stages was covered from cradle-to-grave only for DRAM production. Third, it would be beneficial to include packing process data in future research. Finally, it is worth noting that the setting of global net zero carbon emission target dates will encourage semiconductor manufacturers to implement programs to achieve specific goals. Strategies to meet zero GHG emissions targets, such as low carbon energy and grids, carbon capture, utilization, and storage, should be included in future LCA.

6. Author contributions

Tsai-Chi Kuo: Methodology, investigation, writing, review, and editing. Chen-Yuan Kuo: Writing and editing. Liang Wei Chen: Data investigation

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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